

Discrete Distributions

1 Random Variables of the Discrete Type

Definition 1.1 (Random Variables).

Given a random experiment with sample space S , a function $X: S \rightarrow \mathbb{R}$ that assigns a real number $X(s) = x$ to each $s \in S$ is called a **random variable** (RV).

Definition 1.2 (Discrete Random Variable).

A random variable is said to be of the **discrete type** if its range is *finite* or *countably infinite*.

Definition 1.3 (Probability Mass Function).

A function $f: \mathbb{R} \rightarrow [0, 1]$ is called the **probability mass function** (pmf) of a DRV X with range S if it satisfies:

- $f(x) > 0$ for $x \in S$, and $f(x) = 0$ for $x \notin S$
- $\sum_{x \in S} f(x) = 1$
- $\mathbb{P}(X \in A) = \sum_{x \in A} f(x)$, where $A \subseteq S$

Then S is called the **support** or **space** of X .

Remark. From the third property, we have:

$$\mathbb{P}(X = x) = f(x)$$

Definition 1.4 (Cumulative Distribution Function).

The function $F: \mathbb{R} \rightarrow [0, 1]$ defined by

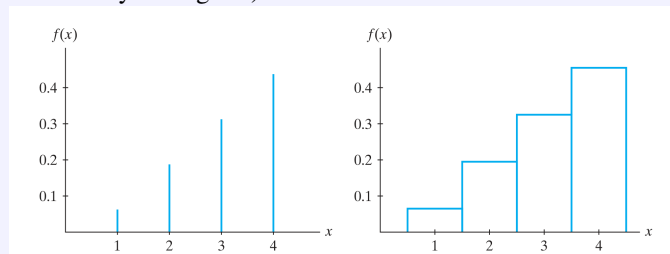
$$F(x) := \mathbb{P}(X \leq x)$$

is called the **cumulative distribution function** (cdf) of X .

Then we have:

$$\mathbb{P}(a < X \leq b) = F(b) - F(a)$$

Definition 1.5 (Line Graph and Probability Histogram).



Definition 1.6 (Uniform Distribution).

An RV X is said to have a **uniform distribution** if its pmf $f(x)$ is constant for all $x \in S$.

2 Mathematical Expectation

Definition 2.1 (Mathematical Expectation).

Let $f(x)$ be the pmf of the DRV X with space S . Suppose g is a function of X , then

$$\mathbb{E}[g(X)] := \sum_{x \in S} g(x)f(x)$$

is called the **mathematical expectation** or **expected value** of g , if the sum exists.

Remark. $g(X)$ gives another RV.

2.1 Properties

Theorem 2.1.

- For any constant c : $\mathbb{E}[c] = c$
- Linearity: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$

Proof (Linearity).

The linearity dose **not** depend on the independence of X and Y . Here gives a proof for the continuous case.

$$\begin{aligned}\mathbb{E}[aX + Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (ax + y)f_{X,Y}(x, y) \, dx \, dy \\ &= a \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dy \, dx + \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dx \, dy \\ &= a \int_{-\infty}^{\infty} x f_X(x) \, dx + \int_{-\infty}^{\infty} y f_Y(y) \, dy \\ &= a\mathbb{E}[X] + \mathbb{E}[Y]\end{aligned}$$

□

3 Special Mathematical Expectations

3.1 Mean & Variance

Definition 3.1 (Mean).

The **mean** of a DRV X (or of its distribution) is defined as

$$\mu := \mathbb{E}[X] = \sum_{x \in S} xf(x)$$

Definition 3.2 (Variance & Standard Deviation).

The **variance** of a DRV X (or of its distribution) is defined as

$$\sigma^2 := \text{Var}(X) := \mathbb{E}[(X - \mu)^2] = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

The **standard deviation** is defined as

$$\sigma := \sqrt{\text{Var}(X)}$$

Remark. Standard deviations are not mathematical expectations.

Remark. $b = \mu$ minimizes $\mathbb{E}[(X - b)^2]$.

$$\text{Var}(X) = \min_b \{ \mathbb{E}[(X - b)^2] \}$$

3.2 Properties of Variances

Theorem 3.1.

Ways to compute the variance:

$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X(X - \mu_X)]$$

Theorem 3.2.

- $\text{Var}(c) = 0$
- $\text{Var}(cX) = c^2 \text{Var}(X)$

3.3 The r th Moment

Definition 3.3 (r th Moment).

Given a positive integer r , the r th moment of an RV X is defined as

$$\mathbb{E}[X^r] = \sum_{x \in S} x^r f(x)$$

The r th moment of the distribution about b :

$$\mathbb{E}[(X - b)^r] = \sum_{x \in S} (x - b)^r f(x)$$

The r th factorial moment:

$$\mathbb{E}[(X)_r] := \mathbb{E}[X(X - 1)(X - 2) \cdots (X - r + 1)]$$

Definition 3.4 (Moment Generating Function).

The **moment generating function** (mgf) of a DRV X is defined as

$$M_X(t) := \mathbb{E}[e^{tX}] = \sum_{x \in S} e^{tx} f(x)$$

if the sum exists and finite for $-h < t < h$.

Theorem 3.3 (Properties of Mgf).

- $M(0) = 1$
- $M^{(r)}(t) = \sum_{x \in S} x^r e^{tx} f(x)$
- $M'(0) = \mathbb{E}[X]$
- $M^{(r)}(0) = \mathbb{E}[X^r]$

Remark. If moments of all order of X exists, then

$$M_X(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!} t^k$$

Theorem 3.4.

2 RVs have the **same mgf** $\iff$ they have the **same probability distribution**, i.e., the **same pmf**

4 The Binomial Distribution

4.1 The Bernoulli Distribution

Bernoulli Experiment The outcomes can be classified in one of two mutually exclusive and exhaustive ways, say either success or failure.

Definition 4.1 (Bernoulli Distribution).

Let X be an RV associated with a *Bernoulli experiment* with the probability of success p . Then $X \in \{0, 1\}$, and the pmf of X is given by

$$f(x) = p^x(1-p)^{1-x}, \quad x \in \{0, 1\}$$

Then we say X has a *Bernoulli distribution* with probability of success p .

Theorem 4.1.

- $\mathbb{E}[X] = p$
- $\text{Var}(X) = p(1-p)$
- $M(t) = e^t p + (1-p)$

Definition 4.2 (Bernoulli Trials).

If a Bernoulli experiment is performed n times

- independently
- with the probability of success remaining the same each

then these n repetitions of the Bernoulli experiment is called n Bernoulli trials.

A random sample of size n from a Bernoulli distribution An observed sequence of n Bernoulli trials that will be an n -tuple of zeros and ones.

4.2 The Binomial Distribution

Description The DRV X is the number of successes in n Bernoulli trials.

Definition 4.3 (Binomial Distribution).

An RV X is said to have a **binomial distribution**, if the pmf is given by

$$f(x) = \binom{n}{p} p^x (1-p)^{n-x} \quad \text{for } x \in \{0, 1, 2, \dots, n\}$$

Then it is denoted by $X \sim B(n, p)$.

Theorem 4.2.

- $M(t) = [e^t p + (1-p)]^n$
- $\mathbb{E}[X] = np$
- $\text{Var}(X) = np(1-p)$

5 The Negative Binomial Distribution

5.1 The Negative Binomial Distribution

Description The DRV X is the number of Bernoulli trials needed to get r successes.

Definition 5.1 (Negative Binomial Distribution).

An RV X is said to have a **negative binomial distribution**, if the pmf is given by

$$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r} \quad \text{for } x \in \{r, r+1, \dots\}$$

Definition 5.2 (Negative Binomial Series).

The negative binomial series is given by Maclaurin's series expansion:

$$(1-w)^{-r} = \sum_{k=0}^{\infty} \binom{k+r-1}{r-1} w^k = \sum_{x=r}^{\infty} \binom{x-1}{r-1} w^{x-r} \quad \text{for } w \in (-1, 1)$$

Theorem 5.1.

- $\mathbb{E}[X] = \frac{r}{p}$
- $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{1-p}{p^2} r$
- $M(t) = \mathbb{E}[e^{tX}] = \left[\frac{pe^t}{1-(1-p)e^t} \right]^r \quad \text{for } (1-p)e^t < 1$

5.2 The Geometric Distribution

Description The DRV X has a *negative binomial distribution* with $r = 1$.

Definition 5.3 (Geometric Distribution).

An RV X is said to have a **geometric distribution** with the probability of success p , if the pmf is given by

$$f(x) = p(1-p)^{x-1} \quad \text{for } x \in \{1, 2, \dots\}$$

Theorem 5.2.

For a positive integer k ,

$$\begin{aligned} \mathbb{P}(X > k) &= \sum_{x=k+1}^{\infty} p(1-p)^{x-1} = (1-p)^k \\ \mathbb{P}(X \leq k) &= 1 - (1-p)^k \end{aligned}$$

6 The Poisson Distribution

Definition 6.1 (Approximate Poisson Process (APP)).

Let the number of occurrences of some event in a given continuous interval be counted. Then we have an APP with parameter $\lambda > 0$ if

- The number of occurrences in non-overlapping subintervals are **independent**.
- The probability of exactly one occurrence in a sufficiently short subinterval of length h is approximately λh .
- The probability of two or more occurrences in a sufficiently short subinterval is essentially 0.

Description Let X be the **number of occurrences** in an APP within a **unit interval**.

Remark. It can be seen as a binomial distribution with $n \rightarrow \infty$ and $p = \frac{\lambda}{n}$, i.e., $\text{Poisson}(\lambda) = \lim_{n \rightarrow \infty} B(n, \frac{\lambda}{n})$.

Definition 6.2 (Poisson Distribution).

An RV X is said to have a **Poisson distribution** with parameter λ , if the pmf is given by

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!} \quad \text{for } x \in \{0, 1, 2, \dots\}$$

Then it is denoted by $X \sim \text{Poisson}(\lambda)$.

Theorem 6.1.

- $M(t) = e^{\lambda(e^t - 1)}$
- $\mathbb{E}[X] = \lambda$
- $\text{Var}(X) = \lambda$

The Poisson distribution works on a **unit interval**, so when considering an interval of length T , we should also treat it as a unit interval. Let X be the number of occurrences. Then $\mathbb{E}[X] = \lambda T$, $X \sim \text{Poisson}(\lambda T)$.

$$f(x) = \frac{(\lambda T)^x e^{-\lambda T}}{x!} \quad \text{for } x \in \{0, 1, 2, \dots\}$$